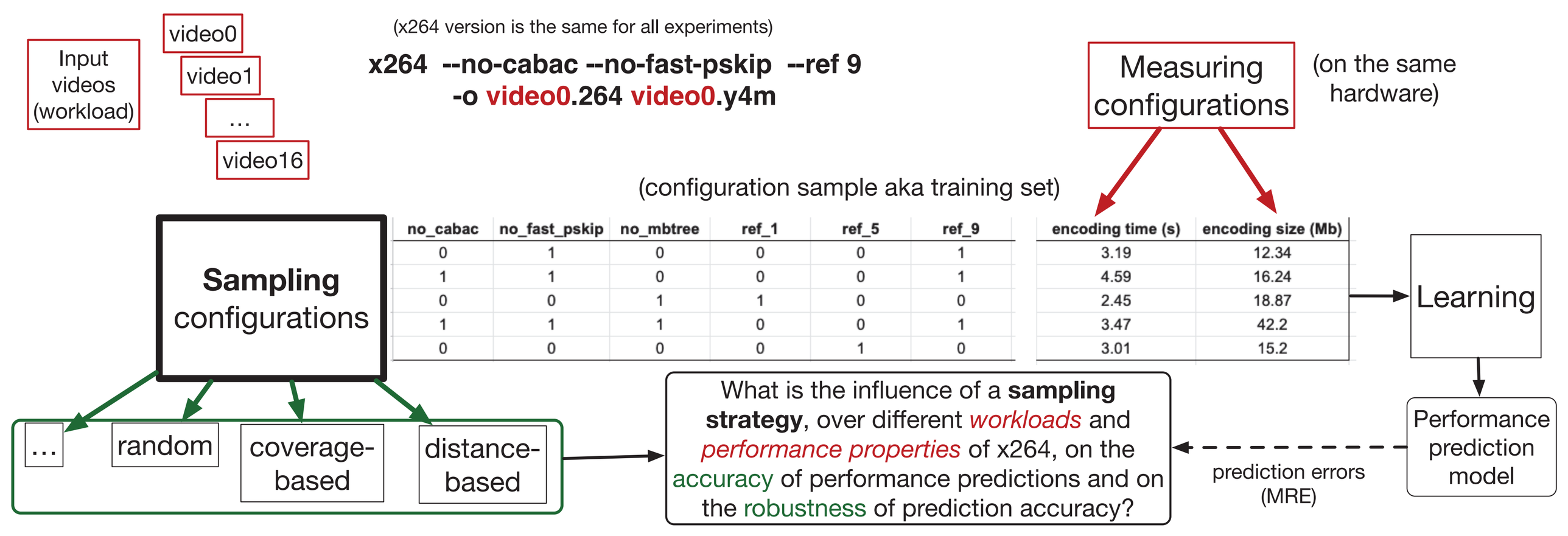
Input videos (workload)
video0
video1
...
video16
(x264 version is the same for all experiments)
x264 --no-cabac --no-fast-pskip --ref 9 -o video0.264 video0.y4m
Measuring configurations
(on the same hardware)
(configuration sample aka training set)
Sampling configurations
no_cabac | no_fast_pskip | no_mbtree | ref_1 | ref_5 | ref_9 | encoding time (s) | encoding size (Mb)
0 | 1 | 0 | 0 | 0 | 1 | 3.19 | 12.34
1 | 1 | 0 | 0 | 0 | 1 | 4.59 | 16.24
0 | 0 | 1 | 1 | 0 | 0 | 2.45 | 18.87
1 | 1 | 1 | 0 | 0 | 1 | 3.47 | 42.2
0 | 0 | 0 | 0 | 1 | 0 | 3.01 | 15.2
Learning
...
random
coverage-based
distance-based
What is the influence of a sampling strategy, over different workloads and performance properties of x264, on the accuracy of performance predictions and on the robustness of prediction accuracy?
prediction errors (MRE)
Performance prediction model